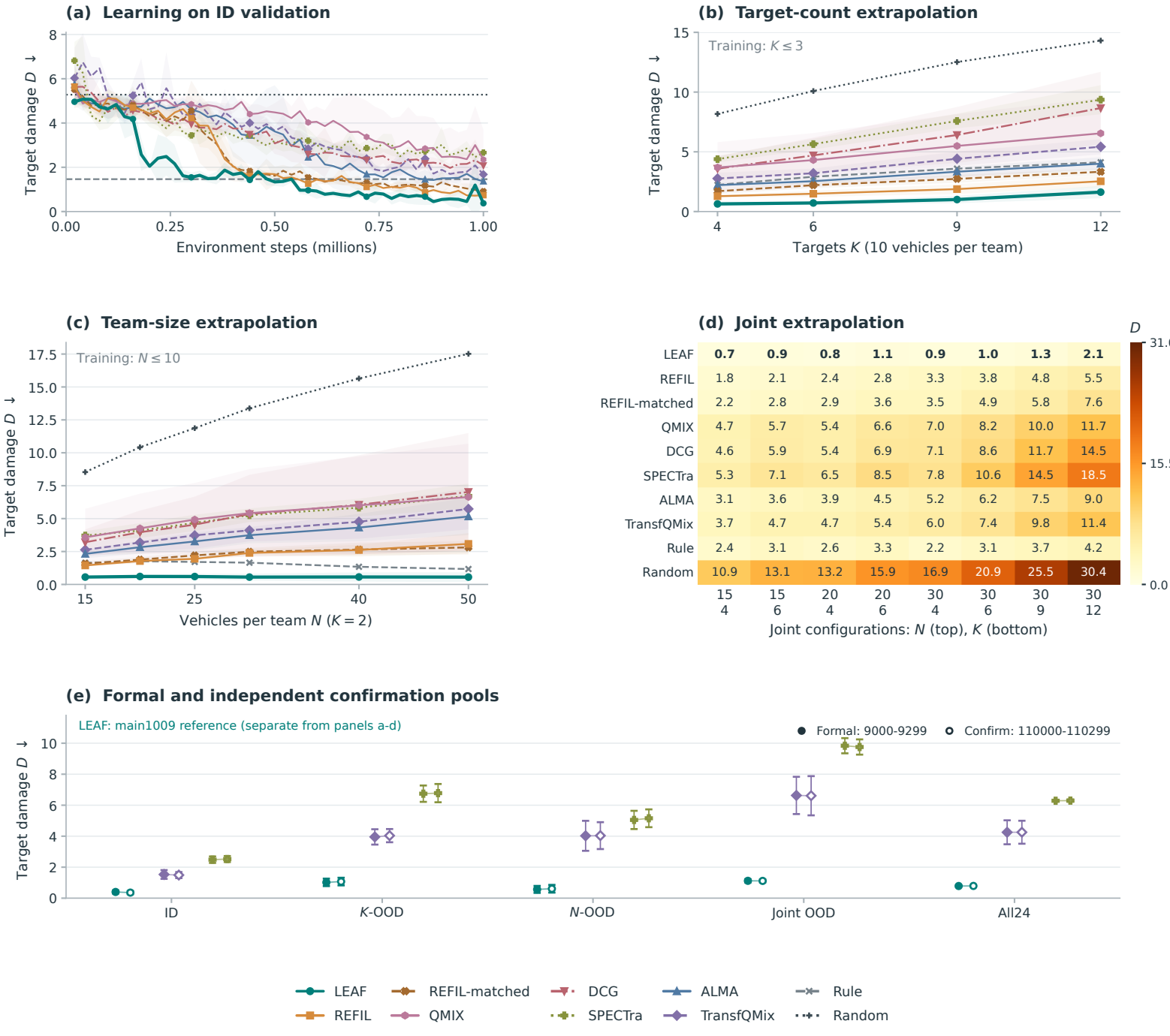

(a) Learning on ID validation
Target damage D ↓
Environment steps (millions)
(b) Target-count extrapolation
Training: K ≤ 3
Target damage D ↓
Targets K (10 vehicles per team)
(c) Team-size extrapolation
Training: N ≤ 10
Target damage D ↓
Vehicles per team N (K = 2)
(d) Joint extrapolation
D
LEAF 0.7 0.9 0.8 1.1 0.9 1.0 1.3 2.1
REFIL 1.8 2.1 2.4 2.8 3.3 3.8 4.8 5.5
REFIL-matched 2.2 2.8 2.9 3.6 3.5 4.9 5.8 7.6
QMIX 4.7 5.7 5.4 6.6 7.0 8.2 10.0 11.7
DCG 4.6 5.9 5.4 6.9 7.1 8.6 11.7 14.5
SPECTra 5.3 7.1 6.5 8.5 7.8 10.6 14.5 18.5
ALMA 3.1 3.6 3.9 4.5 5.2 6.2 7.5 9.0
TransfQMix 3.7 4.7 4.7 5.4 6.0 7.4 9.8 11.4
Rule 2.4 3.1 2.6 3.3 2.2 3.1 3.7 4.2
Random 10.9 13.1 13.2 15.9 16.9 20.9 25.5 30.4
15 15 20 20 30 30 30 30
4 6 4 6 4 6 9 12
Joint configurations: N (top), K (bottom)
(e) Formal and independent confirmation pools
LEAF: main1009 reference (separate from panels a-d)
Formal: 9000-9299
Confirm: 110000-110299
Target damage D ↓
ID
K-OOD
N-OOD
Joint OOD
All24
LEAF
REFIL
REFIL-matched
QMIX
DCG
SPECTra
ALMA
TransfQMix
Rule
Random
Eight trained policies; Rule / Random horizontal lines are ID scene references, not training curves.